



# Deposit type discrimination based on trace elements in sphalerite

Yu-Miao Meng<sup>a</sup>, Xiao-Wen Huang<sup>a,\*</sup>, Ruizhong Hu<sup>a</sup>, Georges Beaudoin<sup>b</sup>, Mei-Fu Zhou<sup>a</sup>, Songning Meng<sup>a</sup>

<sup>a</sup> State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences, Guiyang 550081, China

<sup>b</sup> Département de géologie et de génie géologique, Université Laval, Québec, QC G1V 0A6, Canada

## ARTICLE INFO

### Keywords:

Trace elements  
Sphalerite  
Pb-Zn deposits  
Deposit type discrimination  
Machine learning  
Mineral exploration

## ABSTRACT

Sphalerite can form in different geological environments and is the main ore mineral of most Pb-Zn deposits. Trace elements in sphalerite combined with machine learning (ML) methods have been used for deposit type discrimination. However, the performance of different multivariate and ML methods is not systematically compared. Here more than four thousand laser ablation inductively coupled plasma mass spectrometry data from more than one hundred deposits were compiled and investigated by different multivariate and ML methods. Unsupervised multivariate and ML methods including principal component analysis (PCA) and *t*-distributed stochastic neighbor embedding (*t*-SNE), and supervised multivariate and ML methods including partial least squares-discriminant analysis (PLS-DA), logistic regression (LR), support vector machine (SVM), K-nearest neighbors (KNN), random forest (RF), naïve Bayes (NB), and neural network (NN) are used to classify deposit types. The models are constructed for five deposit types, Mississippi Valley-type (MVT), Volcanogenic Massive Sulfide (VMS), sedimentary exhalative (SEDEX), Epithermal, and Skarn deposits, based on twelve elements in sphalerite, Fe, Pb, Mn, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, and Sb.

The PCA and PLS-DA methods show similar clustering results, where Epithermal, MVT, SEDEX, and Skarn deposits can be separated from other deposit types. The *t*-SNE method is only efficient in discriminating MVT, SEDEX, and Epithermal deposits. The KNN, RF, and NN methods show better performance than LR, SVM, and NB in terms of average accuracy and precision. The PCA, *t*-SNE, and PLS-DA methods have advantages in data visualization, whereas KNN, NN, and RF methods are more suitable for classification tasks. Feature importance analyses of the PLS-DA and RF methods show that the most important discriminant elements in the classification are Ge, Mn, In, Sn, Ga, and Sb. The models based on PCA, PLS-DA, KNN, NN, and RF are used to predict sphalerite samples with uncertain origins. All the models give similar classification results for deposit types, which are consistent with the ore-forming geological setting. The multivariate and ML models, including discrimination diagrams of PCA and PLS-DA, can be used as the first screening for the type of Pb-Zn mineralization in mineral exploration.

## 1. Introduction

Lead-zinc sulfide deposits are major Pb and Zn resources in the world and contain by-product resources of associated metals such as Ag, Ge, Ga, and In (Högl et al., 2007). There are five important types of Pb-Zn deposits, including volcanogenic massive sulfide (VMS), sedimentary exhalative (SEDEX), Mississippi Valley-type (MVT), skarn, and epithermal deposits (Leach et al., 2001; Einaudi et al., 2005). The classification of these deposit types is based on the geological features, ore textures and structures, temperature and composition of ore-forming fluids. For a known deposit, the type is defined based on geological

information for ores. However, how to quickly identify the origin of sphalerite lacking geological information or from overburden sediments is more important in the initial stage of exploration.

Sphalerite is one of the main ore minerals in different types of Pb-Zn deposits. Previous studies have shown that sphalerite can incorporate a wide range of minor and trace elements, including Fe, Ga, Cd, In, Ge, Sn, Ag, Sn, and Cu (Beaudoin, 2000; Belissont et al., 2016; Frenzel et al., 2016; Zhang et al., 2022). The trace element composition of sphalerite has been used to constrain hydrothermal conditions such as temperature and fluid composition. Sphalerite formation temperature is strongly correlated with the trace element content (Stoiber, 1940; Frenzel et al.,

\* Corresponding author.

E-mail address: [huangxiaowen@mail.gyig.ac.cn](mailto:huangxiaowen@mail.gyig.ac.cn) (X.-W. Huang).

<https://doi.org/10.1016/j.oregeorev.2024.105887>

Received 29 November 2023; Received in revised form 5 January 2024; Accepted 10 January 2024

Available online 13 January 2024

0169-1368/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

2016). The absolute element abundances and element ratios of sphalerite can be used to constrain the fluid composition of a specific deposit type (Cook et al., 2009; Ye et al., 2011). Previous studies have demonstrated that the trace element composition of sphalerite can be used to discriminate different types of Pb-Zn deposits. Zhang (1987) proposed a plot of  $\ln(\text{Ga})$  versus  $\ln(\text{In})$  and a ternary plot of  $\text{Se/Te-Ga/In-Zn/Cd}$  to distinguish sphalerite from magmatic-hydrothermal, volcanogenic, modified sedimentary, and sedimentary metamorphic migmatization deposits. Cook et al. (2009) identified important chemical features of sphalerite that can be used to characterize each deposit type. Trace element concentrations in sphalerite from different deposit types are due to different partition coefficients which are controlled by coprecipitated sulfides, temperature, and sulfur fugacity (Cook et al., 2009). Epithermal and some skarn deposits have higher Mn, Co, and In contents than other types of deposits (Cook et al., 2009).

Based on the work of Cook et al. (2009), Ye et al. (2011) further expanded the trace element data of sphalerite and tried to discriminate deposit types using sphalerite chemistry. Sphalerite from skarn deposits is characterized by elevated Co, Mn, and In contents. Sphalerite from sediment-hosted sulfide and VMS deposits is relatively rich in In, Sn, and Ga, whereas that from MVT Pb-Zn deposits is rich in Ge, Cd, Tl, and As. These studies linked sphalerite origin and element contents or ratios to provide a basis for the discrimination between different types of deposits.

Compared to plots of element contents or ratios, multivariate statistical analysis methods such as principal component analysis (PCA) and partial least squares-discriminant analysis (PLS-DA) are better at classifying different deposit types using mineral chemistry (Belissont et al., 2014; Frenzel et al., 2016; Makvandi et al., 2016a; Huang et al., 2019b). Belissont et al. (2014) showed that PCA of Ge, As, Sb, Ga, Ag, Sn, and In in sphalerite can be used to discriminate MVT, vein-type, skarn, VMS, sediment-hosted, and epithermal Pb-Zn deposits, despite overlaps. Similarly, Frenzel et al. (2016) used PCA to distinguish sphalerite from MVT, sediment (clastic)-hosted massive sulfide, VMS, vein-type, and high-temperature hydrothermal (HTHR) replacement deposits or mantos and found that sphalerite from MVT and HTHR deposits can be separated using the first and second principal components. Recently, Sun and Zhou (2022) used decision tree, K-nearest neighbors (KNN), naïve Bayes (NB), random forest (RF), and support vector machine (SVM) methods to classify sphalerite from different types of deposits and considered that decision tree and RF classifiers have high accuracy. Li et al. (2023) used RF to classify the deposits and used *t*-distributed stochastic neighbor embedding (*t*-SNE) to visualize the data and verify the classification results. However, the systematic comparison between multivariate and machine learning (ML) methods is not enough. Therefore, with a larger data set of *in situ* trace elements of sphalerite, the discrimination between different types of Pb-Zn deposits can be further evaluated using both unsupervised and supervised ML methods, including linear and non-linear classification.

In this study, we compiled literature trace element data measured by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) of sphalerite from different types of Pb-Zn deposits worldwide. These data are investigated by unsupervised (PCA, *t*-SNE) and supervised (PLS-DA, KNN, RF, SVM, NB, logistic regression, neural network) multivariate and ML methods to understand which method has better performance and to identify the most important discriminant elements for classification. The models established by these methods are used to predict sphalerite with uncertain origins and the results are compared to the geological information of deposits. The discrimination diagrams based on PCA and PLS-DA can be directly used by other researchers. The identification of sphalerite from regions lacking enough geological information or from overburdened sediments will be helpful for the exploration of Pb-Zn polymetallic deposits.

## 2. Data preparation and machine learning methods

### 2.1. Data collection

Data quality is the first control for the performance of ML models. There are some criteria to select the data: 1) a deposit with multiple origins was not chosen because it is evitable to use a sample with mixed origins. For example, the Baoshan Cu-Pb-Zn deposit includes four mining districts and has both Cu-Mo skarn and Pb-Zn vein mineralization (Zhang et al., 2022). Sphalerite is present in both types of mineralization. 2) a deposit with one or two analyses was not chosen because it has poor representation. For example, both the Herja and Oravita deposits in Romania have only one analysis (George et al., 2016). 3) 12 elements (Fe, Pb, Mn, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, and Sb) are chosen because they have < 40 % of the data below the detection limit or were not analyzed. 4) If more than three of the 12 elements are completely missing for a deposit, these analyses are deleted.

Based on these criteria, we collected 4312 published LA-ICP-MS trace element data in sphalerite from more than 100 Pb-Zn deposits worldwide (Fig. 1). Among them, the number of analyses for sphalerite with known deposit types is 4012. The deposit types are based on literature description and include MVT, VMS, SEDEX, Epithermal, and Skarn. The number of analyses and proportion of different types of mineral deposits are illustrated in Fig. 2. Samples with uncertain genesis (300 analyses) in the literature are used as blind test samples. A summary of the number of analyses for each deposit with known origins is shown in Appendix A Table S1 and a summary of samples with uncertain origins is shown in Appendix A Table S2. The complete chemical data set is available as Appendix A Table S3 and collected in the *MinerGeochem* database (<https://minergeochem.com>).

### 2.2. Data transformation

The data below the detection limit, or not analyzed, were imputed by *robCompositions* package in R (Hron et al., 2010). The appropriate K value is determined by Q-Q plot. The imputed data are further transformed to centered-log ratios (CLR) to overcome the closure problem of concentration calculated to a total of 100 % (Aitchison, 1986; Whitten, 1995). Although CLR transformation does not affect the classification accuracy and precision of NB, KNN, and RF algorithms (Bédard et al., 2022), it will affect multivariate statistical methods such as PCA and PLS-DA (Makvandi et al., 2016b). CLR transformation is thus applied to all multivariate and ML methods in this study.

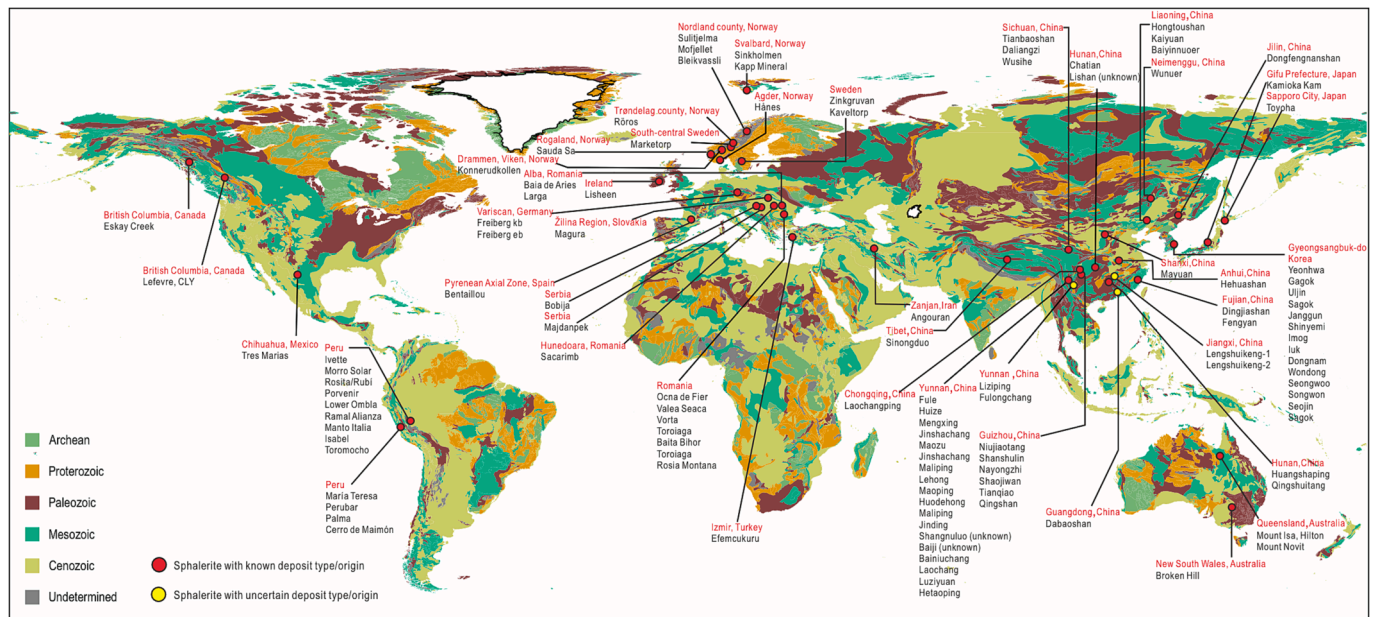
### 2.3. Multivariate and machine learning methods

Both unsupervised and supervised multivariate and ML methods are used to classify deposit types based on trace elements in sphalerite. The workflow for data preparation, transformation, deposit type classification, model construction, and prediction are shown in Fig. 3. For unsupervised multivariate and ML methods, the whole data set (4012 analyses for five deposit types) is used for analysis. For supervised ML methods except for multivariate PLS-DA, 70 % of the data are randomly selected as a training set and the remaining 30 % of data are used as a test set (Fig. 3). PLS-DA was performed on the whole dataset to compare with results of PCA because they are linear regression/classification methods. The model performance was evaluated by the metrics including area under the curve (AUC), classification accuracy (CA), F1 score, precision, and recall. The equations of these metrics have been described in Metz (1978) and listed here for interpretation.

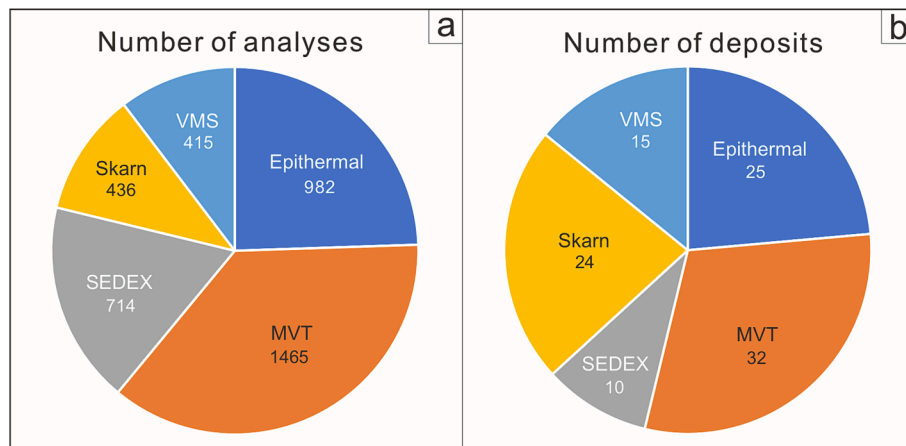
$$P = TP \text{ (true positive)} + FN \text{ (false negative)}$$

$$N = TN \text{ (true negative)} + FP \text{ (false positive)}$$

$$CA = (TP + TN)/(P + N)$$



**Fig. 1.** Summary of the global distribution of all the Pb-Zn deposits discussed in this study. The physical map of the world is from M. Colpron (Geological Survey of Canada).



**Fig. 2.** Pie charts showing the number of analyses (a) and the number of deposits (b) in the model dataset.

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN})$$

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall}/(\text{Precision} + \text{Recall}) = 2 \times \text{TP}/(\text{P} + \text{N} + \text{TP} - \text{TN})$$

The AUC of receiver operating characteristics represents the degree of separability. When  $\text{AUC} = 1$ , the model is perfect and classes are well separated. When  $\text{AUC} = 0.5$ , the model has no discrimination capacity. F1 score combines two competing metrics-precision and recall scores of a model and thus is more useful than CA to evaluate the performance of a model when the data sets are class-imbalanced.

### 2.3.1. Principal component analysis

Principal component analysis is an unsupervised multivariate statistical method. It can reduce the dimensionality of complex element data sets and preserve as much original information as possible. The principal components (e.g., PC1, PC2, PC3) are linear combinations of the original variables (Wold et al., 1987). The importance of principal components decreases because the first principal component captures the maximum variance and then the others. The loading biplots of

elements are used to describe the correlation between elements and the relative contribution of elements for each component. The score biplots are used to show the distribution of samples defined by the element correlation. PCA was performed using *Orange* software (Demšar et al., 2013). All the variables (CLR data of elements) are divided by their standard deviations for data normalization.

The following formulas are used to calculate scores of samples with uncertain origins. The new samples can be plotted in the PC1-PC2 and PC1-PC3 diagrams constructed by PCA.

$$\text{PC1} = (\text{CLR}(a) - \text{average}(a))/\text{stdev}(a) \times \text{loading1}(a) + (\text{CLR}(b) - \text{average}(b))/\text{stdev}(b) \times \text{loading1}(b) + \dots + (\text{CLR}(i) - \text{average}(i))/\text{stdev}(i) \times \text{loading1}(i);$$

$$\text{PC2} = (\text{CLR}(a) - \text{average}(a))/\text{stdev}(a) \times \text{loading2}(a) + (\text{CLR}(b) - \text{average}(b))/\text{stdev}(b) \times \text{loading2}(b) + \dots + (\text{CLR}(i) - \text{average}(i))/\text{stdev}(i) \times \text{loading2}(i);$$

where  $a$ ,  $b$  and  $i$  represent element variables,  $\text{CLR}(i)$  is the centered-log ratio transformed value of element  $i$  for tested sample,  $\text{average}(i)$  and  $\text{stdev}(i)$  are average value and standard deviation of element  $i$  for training sample,  $\text{loading1}(i)$  and  $\text{loading2}(i)$  represent loadings of element  $i$  for training sample in the first and second PCA components.

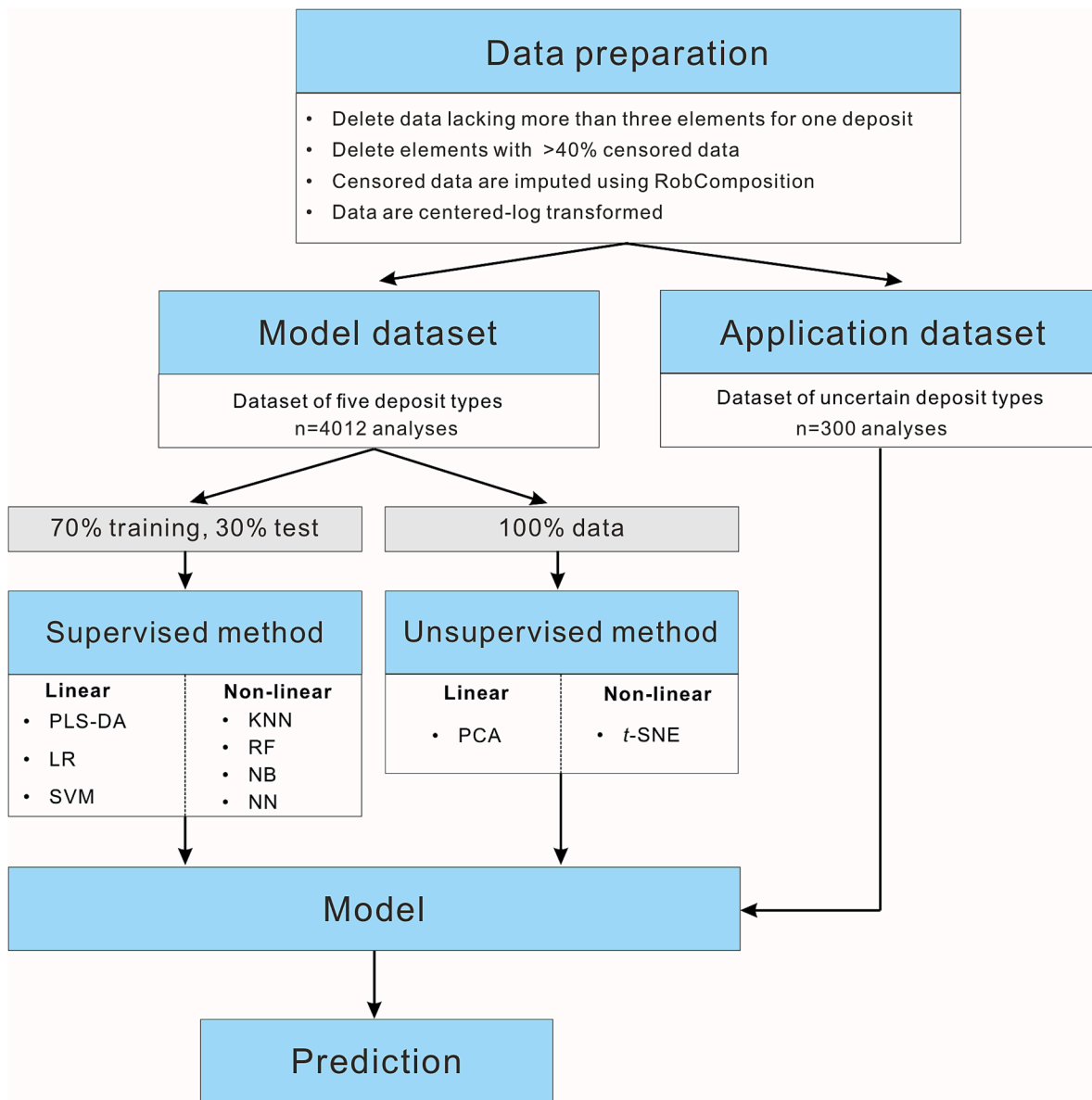


Fig. 3. Flow chart showing the whole procedure for using multivariate and machine learning methods to classify and predict deposit types.

### 2.3.2. *t*-distributed Stochastic neighbor embedding

The *t*-SNE is a non-linear dimensionality reduction algorithm different from linear regression algorithms such as PCA. It can convert high-dimension data sets into a matrix of pair-wise similarities and visualize the results in the two-dimension or three-dimension spaces (Van der Maaten and Hinton, 2008). Because the relationship between trace elements in sphalerite and deposit types is possibly non-linear, the data set was also tested by *t*-SNE. The *t*-SNE was performed on PCA components of 4012 analyses using *Orange* software (Demšar et al., 2013). Eight principal components were used because they account for ~90 % variance of the data set. The model was optimized by using a perplexity value of 25 and an exaggeration value of 1.

### 2.3.3. Partial least squares-discriminant analysis

Partial least squares-discriminant analysis was used to determine whether sphalerite from different deposit types can be distinguished by its trace element composition similar to that described by Makvandi et al., (2016b) and Huang et al., (2019a). PLS-DA was performed using the *plsregress* algorithm in *Matlab* software. Ten-fold cross-validation is used to avoid overfitting of PLS regression. Three types of plots are used

here to interpret PLS-DA results. Loading plots (e.g.,  $q_{w1}$ - $q_{w2}$ ) show the relationship between elements, and labelled groups. Score scatter plots (e.g.,  $t_1$ - $t_2$ ) show the distribution of individual analyses and the relationship among labeled groups. Variable importance on projection (VIP) plot shows the degree of importance of elements in the discrimination where elements with VIP values larger than one are considered to be most important (Eriksson et al., 2013).

For the projection of samples with uncertain origins in the discrimination diagrams established by PLS-DA, we use the following formulas.

$$t_1 = (\text{CLR}(a) - \text{average}(a)) / \text{stdev}(a) \times W_1^*(a) + (\text{CLR}(b) - \text{average}(b)) / \text{stdev}(b) \times W_1^*(b) + \dots + (\text{CLR}(i) - \text{average}(i)) / \text{stdev}(i) \times W_1^*(i);$$

$$t_2 = (\text{CLR}(a) - \text{average}(a)) / \text{stdev}(a) \times W_2^*(a) + (\text{CLR}(b) - \text{average}(b)) / \text{stdev}(b) \times W_2^*(b) + \dots + (\text{CLR}(i) - \text{average}(i)) / \text{stdev}(i) \times W_2^*(i);$$

where  $a$ ,  $b$  and  $i$  represent element variables,  $\text{CLR}(i)$  is the centered-log ratio transformed value of element  $i$  for tested sample,  $\text{average}(i)$  and  $\text{stdev}(i)$  are average value and standard deviation of element  $i$  for training sample,  $W_1^*(i)$  and  $W_2^*(i)$  represent weightings of element  $i$  for training sample in the first and second PLS components.



### 2.3.4. Logistic regression

Logistic regression (LR) is a linear classification algorithm using a sigmoid function. Different from unbounded linear regression, logistic regression is bounded from 0 to 1 (Belyadi and Haghighat, 2021). Moreover, the loss function between linear regression and logistic regression is different. In linear regression, mean squared error or root mean squared error is used as the loss function, whereas maximum likelihood estimation is used in logistic regression (Belyadi and Haghighat, 2021). Logistic regression can be used for binary or multi-class classification. The logistic regression was performed using *Orange* software with default hyperparameters (Demšar et al., 2013). The regularization type was set as Ridge (L2) and the cost strength was set as  $C = 1$  to avoid overfitting.

### 2.3.5. Support vector machine

Support vector machine separates the feature space with an optimal hyperplane that maximizes the distance (margin) between support vectors of the classes (Pisner and Schnyer, 2020). The SVM can be linear or non-linear. For a linear SVM, two types of margins are defined, hard margin and soft margin, based on whether training errors are considered. In the soft-margin method, a penalty factor (cost)  $C$  was introduced to incur a penalty on slack variables (Pisner and Schnyer, 2020), and  $\epsilon$  was used to describe regression loss. In the regression step, linear regression was used in SVM with an  $\epsilon$ -insensitive loss and the accuracy  $a$  can be optimized by appropriate selection of  $C$  and  $\epsilon$ . In the dimensionality reduction step, different kernel models can be used, such as linear, polynomial, radial basis function (RBF), and sigmoid. The SVM was performed using *Orange* software (Demšar et al., 2013), using  $\epsilon$  value of 0.1 and  $C$  value of 1, and an RBF kernel. Default parameters, numerical tolerance of 0.00010 and iteration limit of 100, were used to optimize the model. The RBF kernel is used because it is more efficient to project the non-linearly separable data into linearly separable in a higher dimensional space using a hyperplane.

### 2.3.6. K-nearest Neighbors

K-nearest Neighbors is one of the simplest supervised ML algorithms and can be used for regression and classification. The KNN measures the similarity of instances in the data space and assigns the class label to the majority of the K-nearest patterns (Kramer, 2013). The number of neighbors ( $K$ ) is the most important parameter. The KNN was performed using *Orange* software (Demšar et al., 2013), using  $K = 5$  and Euclidean distance as the metric.

### 2.3.7. Random forest

A random forest is a classifier consisting of an ensemble of decision trees (Breiman, 2001). Each tree is grown from the training data. When individual trees are developed, an arbitrary subset of attributes is constructed by selecting the best attribute for the split. The final result depends on the majority vote from individual trees. The random forest was performed using *Orange* software (Demšar et al., 2013). The number of trees is set as 1000, the number of attributes considered at each split is 3, and the limit depth of individual trees is 15. Feature importance was determined by Rank widget in the *Orange*.

### 2.3.8. Naïve Bayes

Naïve Bayes is a simple learning algorithm based on Bayes' rule by assuming that the attributes are conditionally independent given class (Rish, 2001). Although the independence assumption is often invalid in practice, naïve Bayes shows satisfactory classification accuracy compared to other more complex classifiers (Rish, 2001). The naïve Bayes was performed using *Orange* software.

### 2.3.9. Neural network

Neural network is a multi-layer perceptron algorithm with back-propagation. It is a non-linear function approximator and can be used for both classification and regression. It consists of the input and output

layers, and multiple hidden layers between them (Abiodun et al., 2018). These layers are independent of one another and a specific layer can have an arbitrary number of nodes called bias node (Abiodun et al., 2018). The bias node is usually set as one. When using NN for classification, the input and output nodes will match input features and output classes (Abiodun et al., 2018). The NN was performed using *Orange* software with default parameters (Demšar et al., 2013). The number of neurons in hidden layers is set as 100, the rectified linear unit function (ReLU) is used as an activation function, and the stochastic gradient-based optimizer (Adam) is used as solver for weight optimization. The maximal number of iterations is set as 200.

## 3. Results

### 3.1. Chemical composition of sphalerite from different types of Pb-Zn deposits

The minimum, maximum, median, and average major and trace element contents of sphalerite from five types of Pb-Zn deposits are shown in Fig. 4. Sphalerite from MVT deposits has lower median Fe contents than sphalerite from other deposit types (Fig. 4a). Sphalerite from MVT and SEDEX deposits is characterized by relatively high median Pb and Sb contents (Fig. 4b, l), whereas sphalerite from Epithermal, Skarn and VMS deposits has relatively high median Mn content (Fig. 4c). Cobalt is relatively rich in sphalerite from Skarn deposits (Fig. 4d). Sphalerite from MVT deposits is characterized by lower Cu and In contents (Fig. 4e, j). Gallium is lower in sphalerite from SEDEX and Skarn deposits than other deposit types (Fig. 4f). Germanium tends to be more concentrated in MVT deposits (Fig. 4g). There are no obvious differences in Ag and Cd contents of sphalerite between deposit types (Fig. 4h, i). Sphalerite from Epithermal deposits has higher median Sn contents than other deposit types (Fig. 4k).

### 3.2. Multivariate and ML results

#### 3.2.1. PCA results

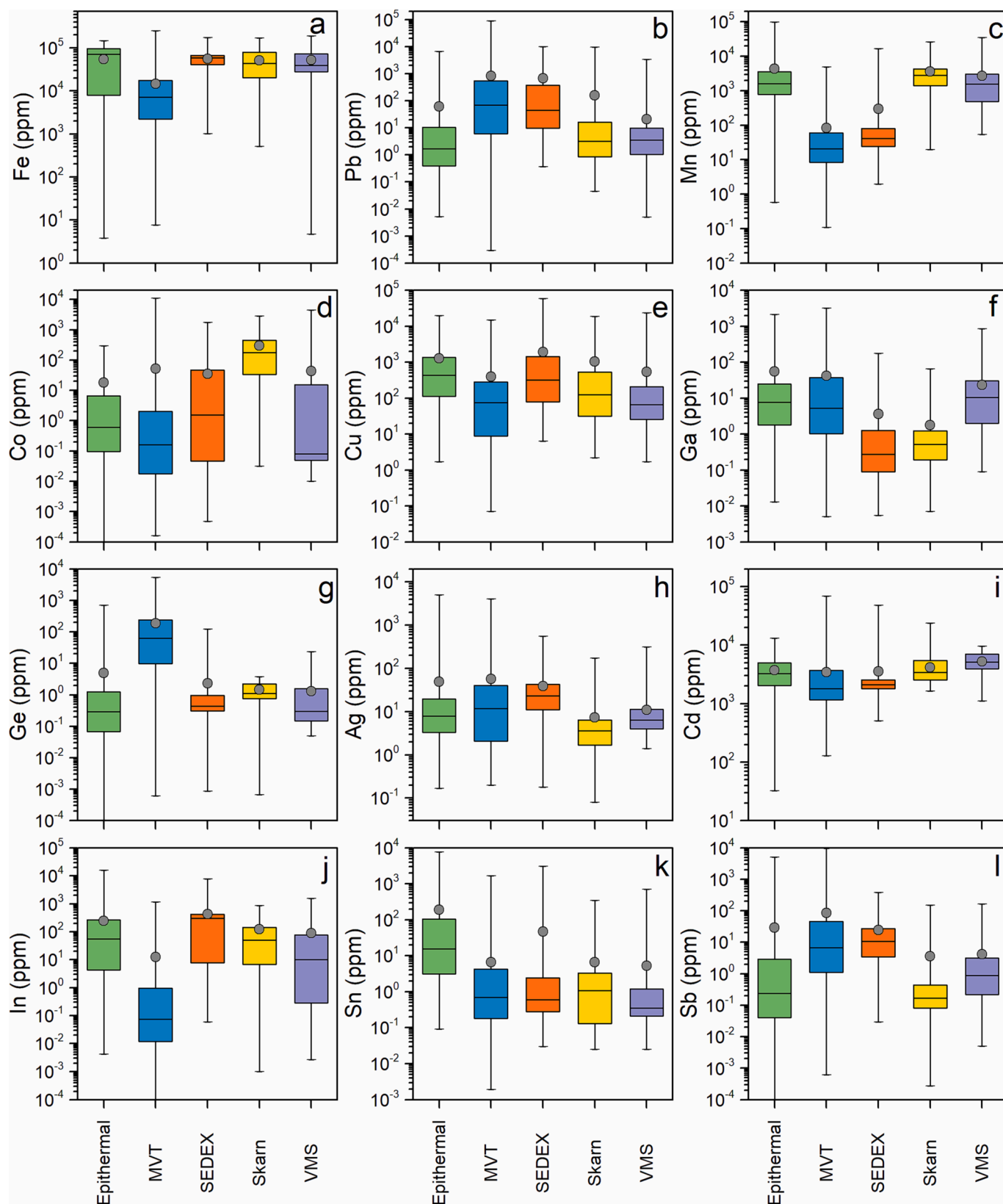
The first three principal components account for 61% of the variance in the data set (Appendix A Fig. S1). In the plot of PC1 and PC2, three deposit types, Epithermal, MVT, and Skarn, can be separated from other deposit types (Fig. 5a). Sphalerite from Epithermal deposits can be discriminated due to a positive correlation with In, Cu, and Sn, whereas that from Skarn deposits can be discriminated because it is positively related to Fe, Co, and Mn (Fig. 5a, Appendix A Fig. S2a). Sphalerite from MVT deposits plots in the positive PC1 region, because they show a positive correlation with Ge, Sb, and Pb (Fig. 5a, Appendix A Fig. S2a). Sphalerite from SEDEX deposits is further discriminated by PC3 (Fig. 5b), due to a positive correlation with Cu, In, and Ag (Appendix A Fig. S2b). Sphalerite from VMS deposits cannot be separated from others by the plots of the first three principal components (Fig. 5a, b).

#### 3.2.2. t-SNE Results

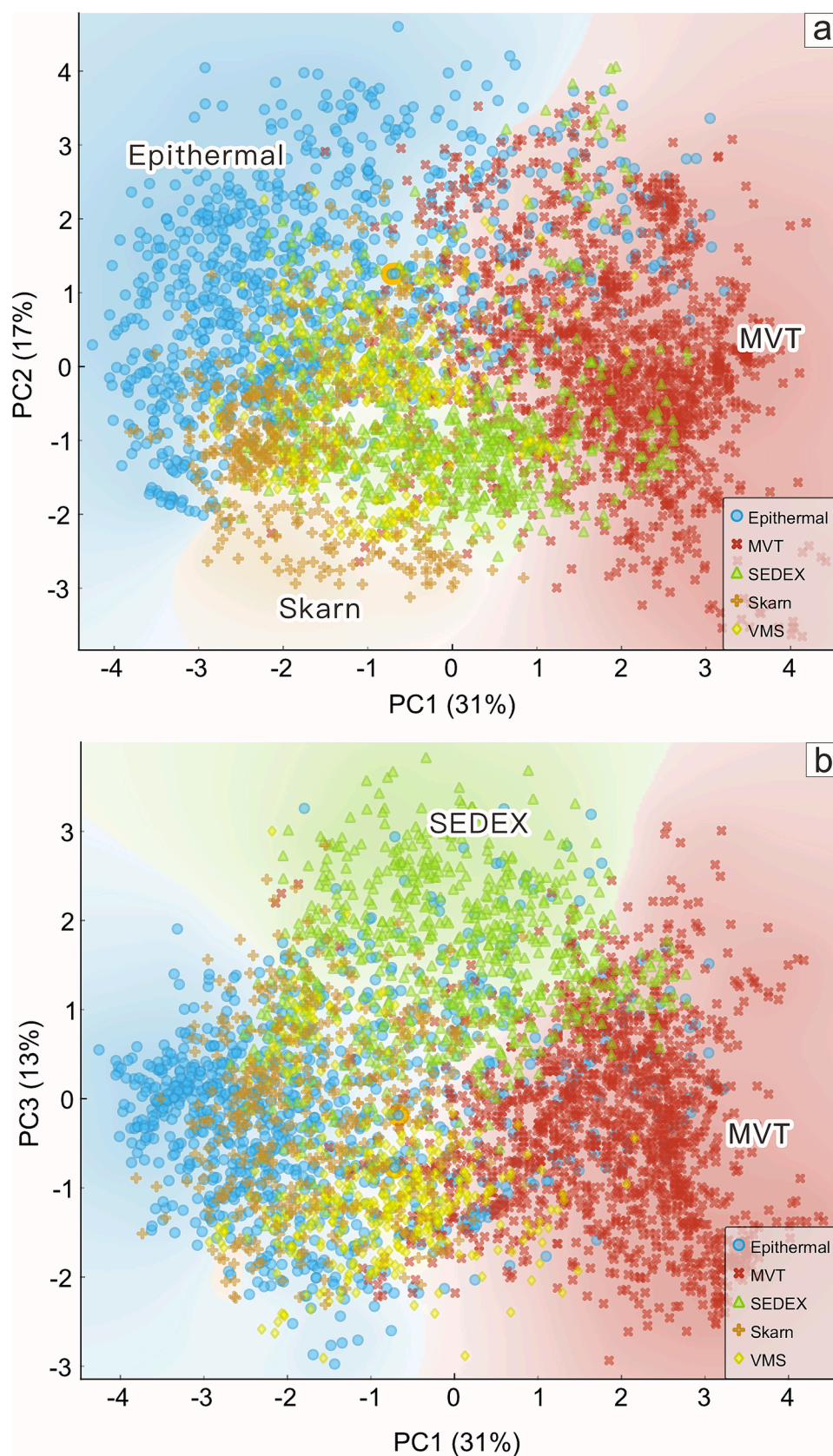
The two-dimensional project results by *t*-SNE for five deposit types are shown in Fig. 6. Similar to PCA results, Epithermal, MVT, and SEDEX deposits can be separated, but Skarn and VMS deposits are largely dispersed (Fig. 6).

#### 3.2.3. PLS-DA results

The PLS-DA results of sphalerite from five types of deposits are shown in Fig. 7. The first three PLS-DA components, accounting for 31 %, 14 %, and 15 % of the variance, respectively, are used for interpretation. Three populations can be identified in the score plot of the first and second PLS-DA components (Fig. 7a). Sphalerite from MVT deposits defines a separated field due to a positive correlation with Pb, Sb, Ag, and Ge, whereas sphalerite from SEDEX deposits is separated because of positively correlated Fe, Pb, and In (Fig. 7a, Appendix A Fig. S3a). Sphalerite from Epithermal deposits is distinguished due to a positive



**Fig. 4.** Box and whisker plots of major and trace elements (Fe, Pb, Mn, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, and Sb) of sphalerite in different types of Pb-Zn deposits analyzed by LA-ICP-MS. The upper and lower margins of the box represent the upper 75 % and lower 25 % of the data. The whiskers represent the minimum and maximum values. Median values are shown as solid black lines and mean values as gray-filled circles. Abbreviations: MVT = Mississippi Valley-type, VMS = volcanogenic massive sulfide, SEDEX = sedimentary exhalative.



**Fig. 5.** Plots of the first principal component versus the second (a) and the third (b) principal components. Numbers in parentheses represent the proportions of variance explained by each principal component. Abbreviations: MVT = Mississippi Valley-type, VMS = volcanogenic massive sulfide, SEDEX = sedimentary exhalative.



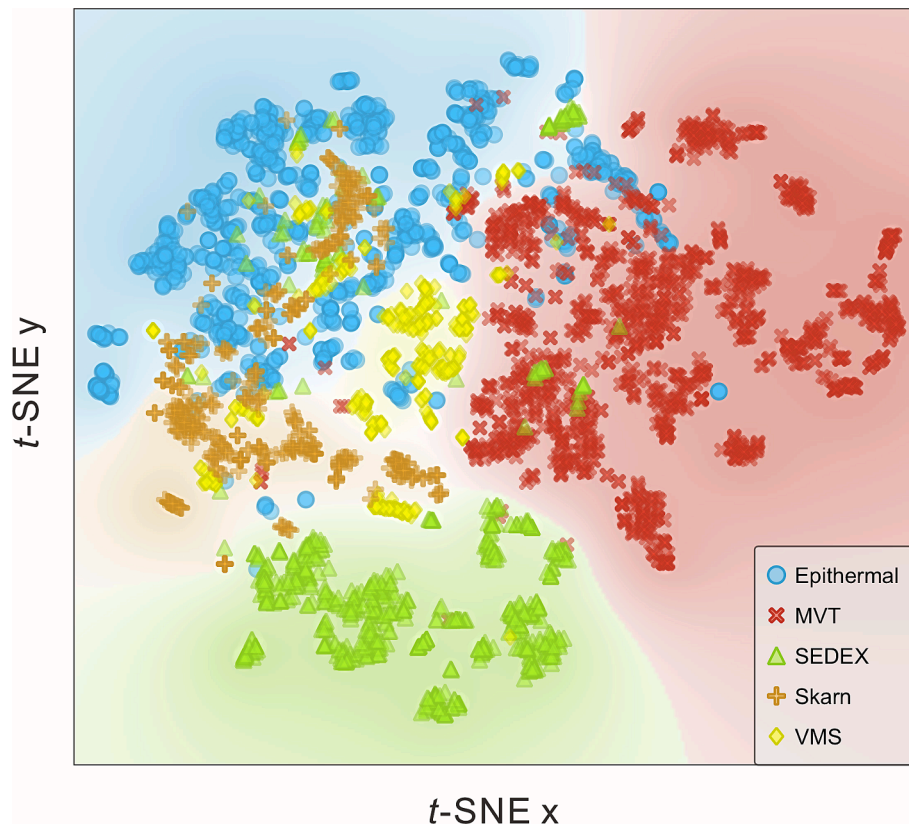


Fig. 6. Two-dimensional t-SNE map showing clusters of different deposit types. The Abbreviations are the same as those in Fig. 5.

correlation with In, Sn, and Mn despite overlapping (Fig. 7a, Appendix A Fig. S3a). Sphalerite from Skarn deposits is further separated by t3, due to a positive correlation with Co and Mn (Fig. 7b, Appendix A Fig. S3b). The VIP diagram shows that Ge, Mn, Sn, In, Sb, Ga and Pb are the most important discriminant elements in the classification (Fig. 8a).

### 3.2.4. LR, SVM, KNN, RF, NB and NN results

The average performance scores of different ML methods are shown in Table 1. The detailed performance scores for a specific class corresponding to a specific ML method are shown in Appendix A Table S4. In terms of F1 score, there is a trend of KNN > RF > NN > LR > NB > SVM (Table 1). Therefore, KNN, RF, and NN algorithms have better performance than other ML algorithms. To know the relative contribution of elements in the classification, feature importance ranking was done by RF. It was shown that the importance of Ge, Mn, In, Sn, Ga, Sb, Fe, Pb, Co, Ag, Cd, and Cu decreases in the RF model (Fig. 8b).

### 3.3. Classification of sphalerite from uncertain settings

A total of 300 analyses from six deposits were used as blind test data sets to predict the types of these deposits using the trained models (Appendix A Table S3). These analyses are not included in the model training. The predicted results are compared with the geological information to verify the reliability of trained models.

#### 3.3.1. Prediction using PCA and PLS-DA

Plotting of samples with uncertain origin in the PCA and PLS-DA models are shown in Figs. 9 and 10, respectively. Sphalerite from the Liziping and Fulongchang Pb-Zn-Ag deposits, SW China (Liu et al., 2021), plots mainly in the field of MVT in the PC1-PC2 and PC1-PC3 diagrams based on the PCA model of five deposit types (Fig. 9a, b). Part of samples from the Shangnuluo deposit (Liu et al., 2021) plot in the MVT field in the PC1-PC2 diagram (Fig. 9a), where nearly all the

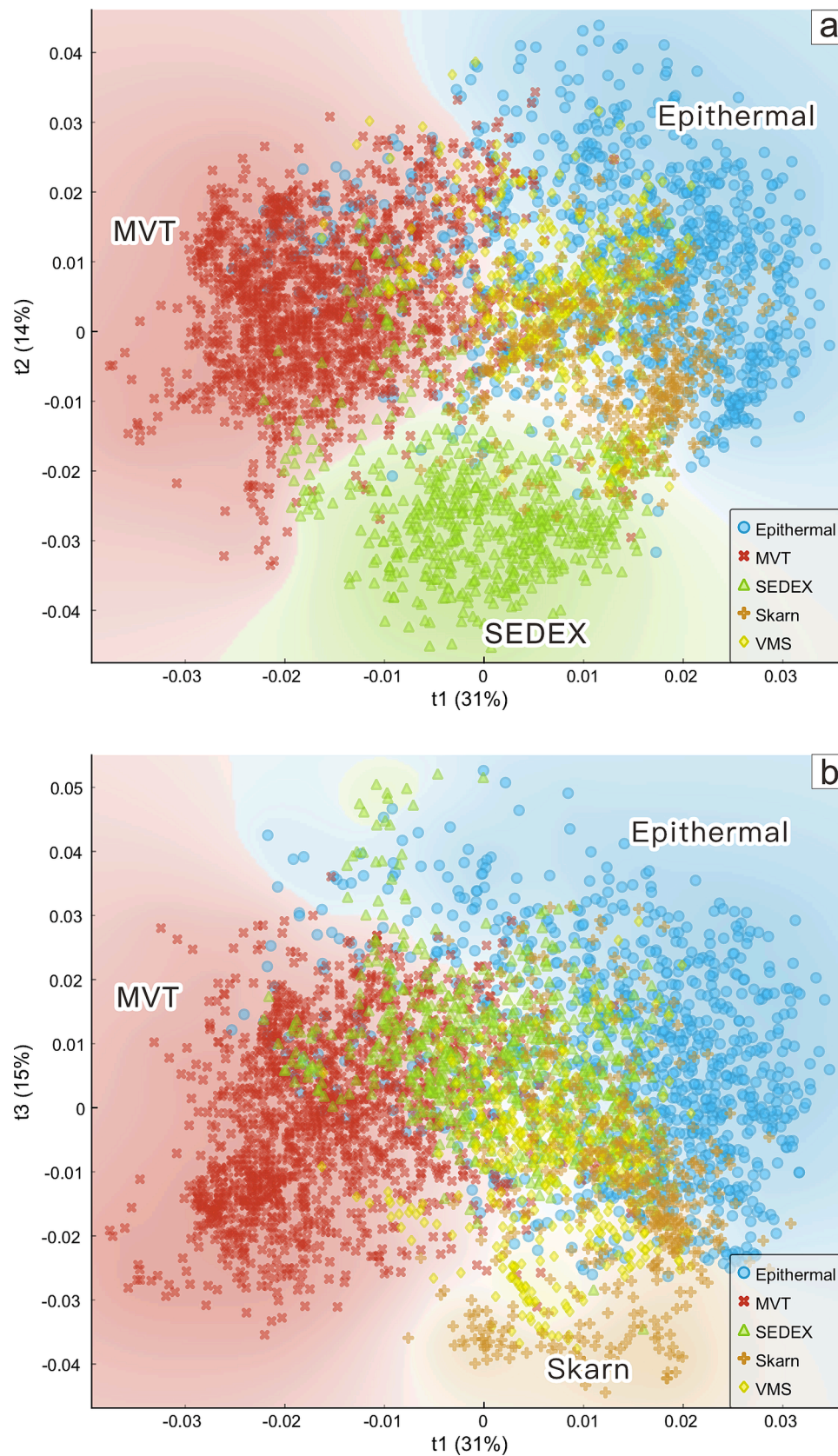
samples plot in the MVT field in the PC1-PC3 diagram (Fig. 9b). Samples from the Baiji deposit (Liu et al., 2021) mainly plot in the MVT field in the PC1-PC2 and PC1-PC3 diagrams (Fig. 9a, b). Samples of Lengshuikeng-1 from the Lengshuikeng deposit (Gong et al., 2019) mainly plot in the Epithermal field in the PC1-PC2 diagram (Fig. 9a), but part of samples is also in the SEDEX field in the PC1-PC3 diagram (Fig. 9b). Samples of Lengshuikeng-2 from the Lengshuikeng deposit (Leng and Qi, 2017) plot in the MVT, Epithermal, SEDEX, and undefined overlapping fields (Fig. 9a, b). Samples from the Lishan deposit (Guo et al., 2020) are mainly distributed in the Epithermal and undefined overlapping fields (Fig. 9a), but some samples are in the SEDEX field (Fig. 9b).

All the samples from the Liziping and Fulongchang deposits and most samples from the Shangnuluo and Baiji deposits plot in the field of MVT in the t1-t2 and t1-t3 diagrams (Fig. 10a, b). Samples from the Lengshuikeng (Lengshuikeng1 and Lengshuikeng2) are dispersed in the MVT, Epithermal, SEDEX, and undefined overlapping fields (Fig. 10a, b). Samples from the Lishan deposit plot in the Epithermal field and undefined overlapping fields in the t1-t2 diagram (Fig. 10a), but they plot in the undefined overlapping fields in the t1-t3 diagram (Fig. 10b).

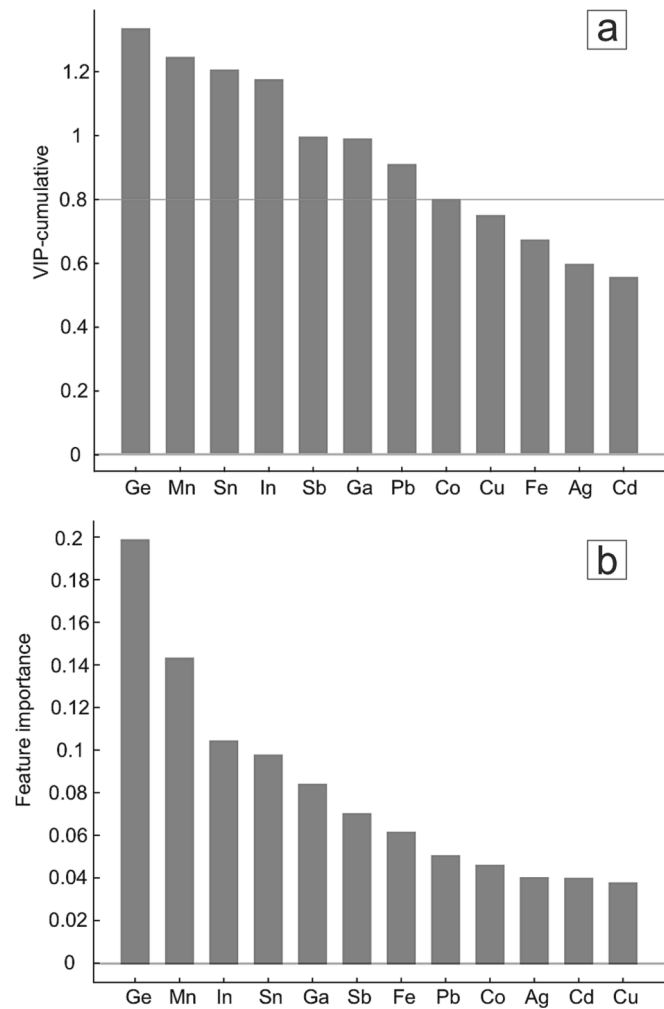
#### 3.3.2. Prediction using KNN, NN, and RF algorithms

Table 2 shows the prediction results of sphalerite with uncertain origin. The proportion of analyses identified as MVT for the Liziping, Fulongchang, Shangnuluo, and Baiji deposits, SW China (Liu et al., 2021) is 79.1 %–89.6 %, 79.2 %–91.7 %, 70.8 %–93.8 %, and 56.5 %–78.3 %, respectively (Table 2). These deposits are thus can be considered as mainly MVT deposits based on the discrimination results. Sphalerite samples from the stratabound Pb-Zn mineralization of the Lengshuikeng deposit, SE China (Lengshuikeng1 and Lengshuikeng2) are identified as having multiple origins. 41.7 % of the Lengshuikeng1 samples (Gong et al., 2019) are identified as SEDEX, whereas 16.7 %–45.8 % are identified as Epithermal (Table 2). Minor amounts of





**Fig. 7.** PLS-DA results of trace elements in sphalerite from five deposit types. Score plots of first versus second PLS components (a) and first versus third PLS components (b) showing the compositional range of samples from different types of deposits in the latent variable space defined by  $qw^*1$ - $qw^*2$  and  $qw^*1$ - $qw^*3$  in [Appendix A Fig. S3](#), respectively. Numbers in parentheses represent the proportions of variance explained by each PLS component. Abbreviations are the same as those in [Fig. 5](#).



**Fig. 8.** (a) The cumulative VIP plot based on PLS-DA model showing the importance of compositional variables in the classification of samples. Blue line represents the VIP value of 0.8. Elements with VIP values higher than 0.8 are considered to be important in the classification. (b) Feature importance calculated by the random forest model.

**Table 1**

Summary of average performance scores of models based on five deposit types.

| Model               | AUC   | CA    | F1    | Precision | Recall |
|---------------------|-------|-------|-------|-----------|--------|
| kNN                 | 0.991 | 0.938 | 0.939 | 0.939     | 0.938  |
| Logistic Regression | 0.966 | 0.824 | 0.823 | 0.823     | 0.824  |
| Naive Bayes         | 0.947 | 0.796 | 0.795 | 0.798     | 0.796  |
| Neural Network      | 0.992 | 0.924 | 0.925 | 0.925     | 0.924  |
| Random Forest       | 0.993 | 0.931 | 0.931 | 0.932     | 0.931  |
| SVM                 | 0.937 | 0.723 | 0.72  | 0.724     | 0.723  |

Lengshuikeng1 samples (12.5 %–29.5 %) are identified as Skarn. The 33.3 %–42.9 % of the Lengshuikeng2 samples (Leng and Qi, 2017) are identified as MVT, whereas 33.3 %–38.1 % are identified as SEDEX (Table 2). The 9.5 %–14.3 % of the Lengshuikeng2 samples are identified as Epithermal. For the Lishan Pb–Zn–Cu deposit in the northeastern Hunan Province, China (Guo et al., 2020), 46.4 %–58 % of analyses are identified as Skarn (Table 2).

## 4. Discussion

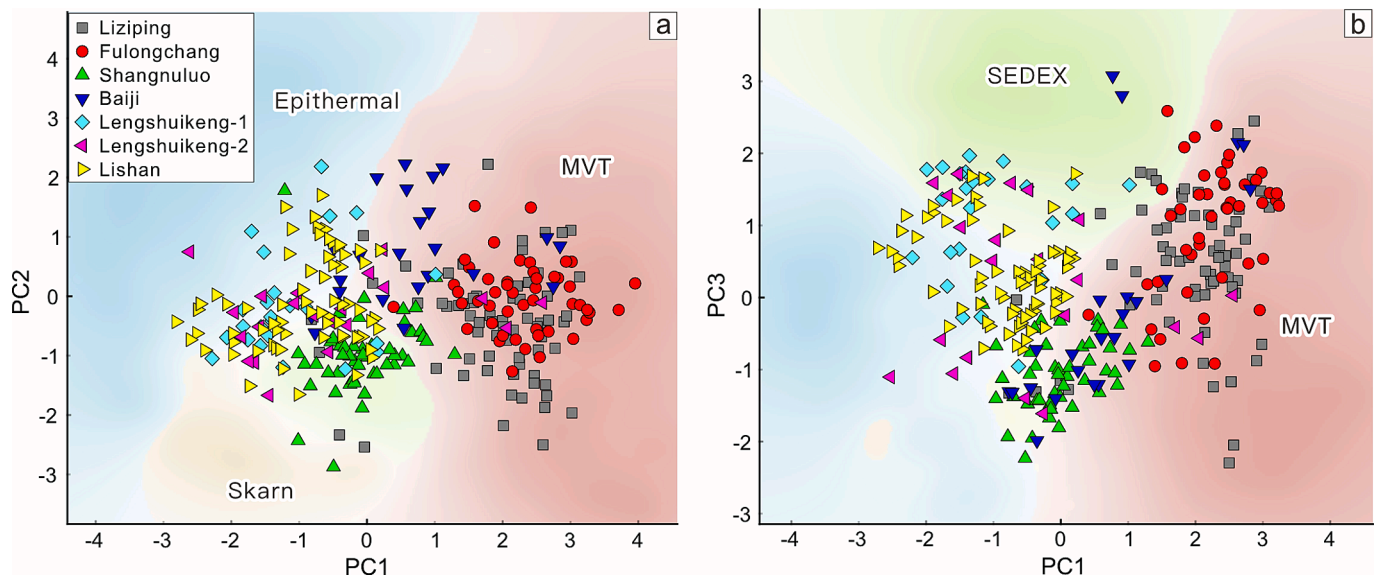
### 4.1. Discrimination between different types of Pb–Zn deposits

A series of diagrams based on the trace element composition of sphalerite have been used to distinguish different origins of sphalerite. Zhang (1987) analyzed trace elements in sphalerite from nearly 30 deposits in China, and proposed plots of  $\ln(\text{Ga})$ – $\ln(\text{In})$  and  $\text{Se}/\text{Te}$ – $\text{Ga}/\text{In}$ – $\text{Zn}/\text{Cd}$  to distinguish sphalerite from magmatic-hydrothermal, volcanogenic, modified sedimentary, and sedimentary metamorphic migmatization deposits. Sphalerite with  $\text{Ga}/\text{In}$  ratios of  $>1$  was shown to characterize modified sedimentary in origin, whereas that with  $\text{Ga}/\text{In}$  ratios of  $<1$  was considered to be magmatic-hydrothermal in origin. Sphalerite from VMS deposits partly overlaps the field of modified sedimentary and magmatic-hydrothermal deposits in both diagrams of Zhang (1987). Belissont et al. (2014) used PCA to discriminate sphalerite from epithermal, MVT, SEDEX, sediment-hosted, skarn, vein-type, and VMS deposits based on seven trace elements, Ga, Ge, As, Ag, In, Sn, and Sb. Vein-type deposits show two compositional populations due to the various chemistry of sphalerite (Belissont et al., 2014). The chemical diversity is also observed for epithermal and sediment-hosted deposits (Belissont et al., 2014). Overlap of sphalerite composition between epithermal, SEDEX, and VMS, and between MVT and sediment-hosted deposits are obvious (Belissont et al., 2014). Principal component analysis of Ag, Cd, Co, Cu, Fe, Ga, Ge, In, and Mn in sphalerite show that MVT deposits can be separated from sediment (clastic)-hosted massive sulfide, VMS, vein-type, and HTHR deposits using first and second principal components due to relative enrichment of Ga and Ge (Frenzel et al., 2016). However, sphalerite samples from sediment (clastic)-hosted massive sulfide, VMS, Vein-type, and HTHR deposits cannot be separated from each other based on this limited suite of elements (Frenzel et al., 2016). Therefore, diagrams based on element contents or ratios (Zhang, 1987) and PCA (Belissont et al., 2014; Frenzel et al., 2016) cannot effectively discriminate sphalerite from different types of deposits.

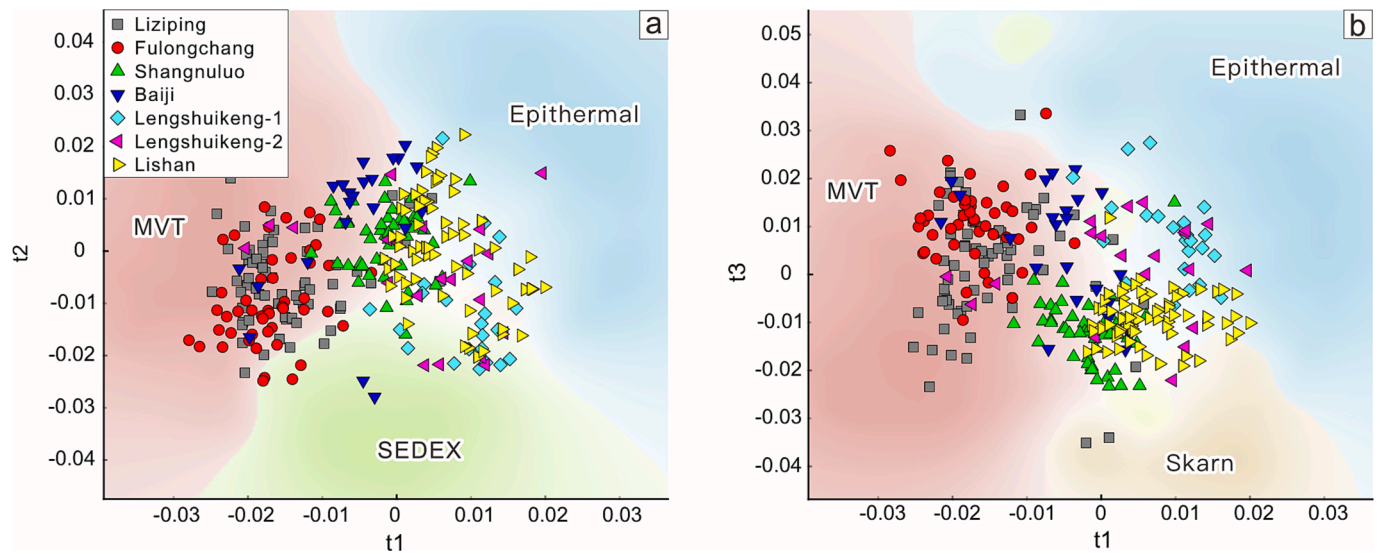
Principal component analysis of 4012 analyses using 12 elements yields better discrimination. Epithermal, Skarn, MVT, and SEDEX deposits are separated from each other (Fig. 5a, b).  $t$ -SNE model shows similar results with PCA (Fig. 6), indicating that the model performance of unsupervised methods is independent of linear or non-linear regression. Compared to the unsupervised PCA and  $t$ -SNE, the supervised PLS-DA didn't show better performance in the discrimination of different origins of sphalerite (Fig. 7). Similarly, Epithermal, Skarn, MVT, and SEDEX deposits are separated from each other by the first three PLS components (Fig. 7). The supervised ML methods such as KNN, NN, and RF have a stronger ability in classifying different types of deposits. Deposit types such as VMS cannot be discriminated by PCA and PLS-DA, but can be discriminated by KNN, NN, and RF. This indicates non-linear, supervised ML methods are more suitable for classification tasks. Although PLS-DA and RF models show different performance, both of them identify that Ge, Mn, In, Sn, Ga, and Sb are the most important discriminant elements (Fig. 8). The results are consistent with those of Li et al. (2023) using RF model. In summary, PCA,  $t$ -SNE and PLS-DA methods have the advantages of data visualization, whereas KNN, NN, and RF algorithms are superior to them in terms of classification ability.

### 4.2. Application in mineral exploration

Chemistry of indicator minerals such as magnetite, pyrite, apatite, chlorite, and scheelite has been used in mineral exploration (Bouzari et al., 2016; Makvandi et al., 2016b; Mukherjee and Large, 2017; Pisiak et al., 2017; Cooke et al., 2020; Sciuba et al., 2021; Miranda et al., 2022). The variations in trace elements with the depth or distance from the mineralization center provide geochemical vectors in exploration. However, the application of sphalerite chemistry in mineral exploration



**Fig. 9.** Deposit type prediction of sphalerite with uncertain origins using the PCA models of Fig. 5. The formulas in a and b are:  $PC1 = (CLR(Fe) - 2.8515) \times (-0.2898) + (CLR(Pb) + 0.2362) \times 0.3248 + (CLR(Mn) - 0.8763) \times (-0.3353) + (CLR(Co) + 1.3747) \times (-0.1743) + (CLR(Cu) - 0.7874) \times (-0.2258) + (CLR(Ga) + 0.8775) \times 0.1154 + (CLR(Ge) + 0.9411) \times 0.3072 + (CLR(Ag) + 0.3307) \times 0.3463 + (CLR(Cd) - 2.0859) \times (-0.1531) + (CLR(In) + 0.7340) \times (-0.2674) + (CLR(Sn) + 1.0757) \times (-0.1465) + (CLR(Sb) + 1.0311) \times 0.3772$ .  $PC2 = (CLR(Fe) - 2.8515) \times (-0.5487) + (CLR(Pb) + 0.2362) \times (-0.1252) + (CLR(Mn) - 0.8763) \times (-0.0575) + (CLR(Co) + 1.3747) \times (-0.2692) + (CLR(Cu) - 0.7874) \times 0.3169 + (CLR(Ga) + 0.8775) \times 0.4490 + (CLR(Ge) + 0.9411) \times (-0.0543) + (CLR(Ag) + 0.3307) \times (-0.01988) + (CLR(Cd) - 2.0859) \times (-0.5731) + (CLR(In) + 0.7340) \times 0.0677 + (CLR(Sn) + 1.0757) \times 0.6067 + (CLR(Sb) + 1.0311) \times 0.0083$ .  $PC3 = (CLR(Fe) - 2.8515) \times (-0.1272) + (CLR(Pb) + 0.2362) \times 0.1534 + (CLR(Mn) - 0.8763) \times (-0.2953) + (CLR(Co) + 1.3747) \times 0.0990 + (CLR(Cu) - 0.7874) \times 0.5538 + (CLR(Ga) + 0.8775) \times (-0.4598) + (CLR(Ge) + 0.9411) \times (-0.1833) + (CLR(Ag) + 0.3307) \times 0.3162 + (CLR(Cd) - 2.0859) \times (-1.0476) + (CLR(In) + 0.7340) \times 0.1898 + (CLR(Sn) + 1.0757) \times (-0.0618) + (CLR(Sb) + 1.0311) \times 0.1381$ .



**Fig. 10.** Deposit type prediction of sphalerite with uncertain origins using the PLS-DA models of Fig. 7. The formulas in a and b are:  $t1 = (CLR(Fe) - 2.8515) \times 0.0017 + (CLR(Pb) + 0.2362) \times (-0.0022) + (CLR(Mn) - 0.8763) \times 0.0032 + (CLR(Co) + 1.3747) \times 0.0011 + (CLR(Cu) - 0.7874) \times 0.0020 + (CLR(Ga) + 0.8775) \times (-0.0009) + (CLR(Ge) + 0.9411) \times (-0.0033) + (CLR(Ag) + 0.3307) \times (-0.0022) + (CLR(Cd) - 2.0859) \times 0.00001 + (CLR(In) + 0.7340) \times 0.0022 + (CLR(Sn) + 1.0757) \times 0.0019 + (CLR(Sb) + 1.0311) \times (-0.0027)$ .  $t2 = (CLR(Fe) - 2.8515) \times (-0.0041) + (CLR(Pb) + 0.2362) \times (-0.0027) + (CLR(Mn) - 0.8763) \times 0.0032 + (CLR(Co) + 1.3747) \times (-0.0013) + (CLR(Cu) - 0.7874) \times (-0.0017) + (CLR(Ga) + 0.8775) \times 0.0073 + (CLR(Ge) + 0.9411) \times 0.0022 + (CLR(Ag) + 0.3307) \times (-0.0025) + (CLR(Cd) - 2.0859) \times 0.0058 + (CLR(In) + 0.7340) \times (-0.0028) + (CLR(Sn) + 1.0757) \times 0.0049 + (CLR(Sb) + 1.0311) \times (-0.0025)$ .  $t3 = (CLR(Fe) - 2.8515) \times (-0.0030) + (CLR(Pb) + 0.2362) \times 0.0004 + (CLR(Mn) - 0.8763) \times (-0.0039) + (CLR(Co) + 1.3747) \times (-0.0037) + (CLR(Cu) - 0.7874) \times 0.0052 + (CLR(Ga) + 0.8775) \times (-0.0002) + (CLR(Ge) + 0.9411) \times (-0.0029) + (CLR(Ag) + 0.3307) \times 0.0046 + (CLR(Cd) - 2.0859) \times (-0.0070) + (CLR(In) + 0.7340) \times 0.0021 + (CLR(Sn) + 1.0757) \times 0.0072 + (CLR(Sb) + 1.0311) \times 0.0022$ .

is rarely reported due to a lack of data and discriminant criteria.

The first step for the indicator mineral methods is to identify the type of mineral deposit. Sphalerite with uncertain origins is used here to test the discrimination ability of different multivariate and ML methods (Figs. 5 and 7, Table 2). Sphalerite from the Liziping and Fulongchang

Pb-Zn-Ag deposits plots in the MVT field based on PCA and PLS-DA models (Figs. 9 and 10). The Liziping, Fulongchang, Shangnuluo, and Baiji deposits are predicted as MVT by the KNN, NN, and RF models (Table 2). The discrimination results of these four deposits are consistent with the MVT-like description of them (Liu et al., 2021). The



**Table 2**  
Prediction deposit types of sphalerite samples with uncertain origins.

| Deposit name   | Predicted deposit type | Model based on five deposit types |                |               |
|----------------|------------------------|-----------------------------------|----------------|---------------|
|                |                        | KNN                               | Neural network | Random forest |
| Liziping       | Epithermal MVT         | 0                                 | 2.99 %         | 0             |
|                |                        | <b>79.1 %</b>                     | <b>86.6 %</b>  | <b>89.6 %</b> |
|                | SEDEX                  | 11.9 %                            | 6.0 %          | 4.5 %         |
|                |                        |                                   |                |               |
| Fulongchang    | Skarn                  | 6.0 %                             | 3.0 %          | 4.5 %         |
|                | VMS                    | 3.0 %                             | 1.5 %          | 1.5 %         |
|                | Epithermal MVT         | 4.2 %                             | 2.1 %          | 4.2 %         |
|                |                        | <b>87.5 %</b>                     | <b>79.2 %</b>  | <b>91.7 %</b> |
|                | SEDEX                  | 8.3 %                             | 18.8 %         | 4.2 %         |
|                |                        |                                   |                |               |
| Shangnuluo     | Skarn                  | 0                                 | 0              | 0             |
|                | VMS                    | 0                                 | 0              | 0             |
|                | Epithermal MVT         | 8.3 %                             | 0              | 2.1 %         |
|                |                        | <b>70.8 %</b>                     | <b>93.8 %</b>  | <b>81.3 %</b> |
|                | SEDEX                  | 10.4 %                            | 4.2 %          | 16.7 %        |
|                |                        |                                   |                |               |
| Baiji          | Skarn                  | 0                                 | 0              | 0             |
|                | VMS                    | 10.4 %                            | 2.1 %          | 0.0 %         |
|                | Epithermal MVT         | 34.8 %                            | 17.4 %         | 21.7 %        |
|                |                        | <b>56.5 %</b>                     | <b>73.9 %</b>  | <b>78.3 %</b> |
|                | SEDEX                  | 8.7 %                             | 0              | 0             |
|                |                        |                                   |                |               |
| Lengshuikeng-1 | Skarn                  | 0                                 | 0              | 0             |
|                | VMS                    | 0                                 | 8.7 %          | 0             |
|                | Epithermal MVT         | 16.7 %                            | 33.3 %         | 45.8 %        |
|                |                        |                                   |                |               |
|                | SEDEX                  | 4.2 %                             | 0              | 0             |
|                |                        | <b>41.7 %</b>                     | <b>41.7 %</b>  | <b>41.7 %</b> |
| Lengshuikeng-2 | Skarn                  | 29.2 %                            | 25.0 %         | 12.5 %        |
|                | VMS                    | 8.3 %                             | 0              | 0             |
|                | Epithermal MVT         | 9.5 %                             | 14.3 %         | 14.3 %        |
|                |                        | <b>33.3 %</b>                     | <b>38.1 %</b>  | <b>42.9 %</b> |
|                | SEDEX                  | 38.1 %                            | 33.3 %         | 33.3 %        |
|                |                        |                                   |                |               |
| Lishan         | Skarn                  | 9.5 %                             | 4.8 %          | 0.0 %         |
|                | VMS                    | 9.5 %                             | 9.5 %          | 9.5 %         |
|                | Epithermal MVT         | 18.8 %                            | 0.0 %          | 7.3 %         |
|                |                        |                                   |                |               |
|                | SEDEX                  | 8.7 %                             | 1.5 %          | 27.5 %        |
|                |                        | 4.4 %                             | 4.4 %          | 5.8 %         |
|                | Skarn                  | <b>58.0 %</b>                     | <b>46.4 %</b>  | <b>58.0 %</b> |
|                | VMS                    | 10.1 %                            | 47.8 %         | 1.5 %         |

Note: Numbers in bold represent the proportions of major deposit types.

Lengshuikeng deposit in Jiangxi Province (SE China) includes two types of sulfide mineralization, one disseminated, veinlet, and vein-type sulfides hosted by granitic porphyry and the other stratabound sulfide mineralization hosted in tuff (Leng and Qi, 2017). Samples of Lengshuikeng1 (Gong et al., 2019), described as stratabound Pb-Zn mineralization, are mainly identified as SEDEX, Epithermal, and Skarn origins, whereas samples of Lengshuikeng2 from the stratabound mineralization (Leng and Qi, 2017), are mainly identified as MVT, SEDEX, and Epithermal origins (Figs. 9 and 10, Table 2). The multiple origins of the Lengshuikeng stratabound Pb-Zn mineralization are consistent with the hypothesis that this type of mineralization was originally SEDEX and was overprinted by hydrothermal fluids related to a porphyry magma system (Wang et al., 2014; Li and Jiang, 2016). Samples from the Lishan Pb-Zn-Cu deposit in the NE Hunan are identified as Skarn origin, consistent with a magmatic-hydrothermal origin of the deposit (Guo et al., 2020; Yu et al., 2021).

The results of discrimination of sphalerite with uncertain origin, using the proposed models, demonstrate that sphalerite chemistry can be used to identify broad classes of deposit types. The discrimination of sphalerite could be useful for first screening during mineral exploration. For sphalerite lacking geological information, we can analyze the trace element composition of sphalerite and plot the data on PCA and PLS-DA diagrams, or put them in the ML models such as KNN, NN, and RF to quickly identify the possible deposit types. The results can be used to evaluate the mineralization potential of Pb-Zn deposits. Moreover, different types of Pb-Zn deposits commonly host critical metals such as In, Ga, Ge, and Co. Therefore, identification of the sphalerite-bearing deposit types is also meaningful for resource evaluation of critical metals.

## 5. Conclusions

More than four thousand laser ablation inductively coupled plasma mass spectrometry trace element data of sphalerite were compiled from > 100 Pb-Zn deposits. These data cover sphalerite from different types of deposits or settings, i.e., Epithermal, MVT, SEDEX, Skarn, and VMS. Unsupervised (PCA, t-SNE) and supervised (PLS-DA, KNN, RF, LR, NN, SVM, NB) multivariate and ML methods are used to classify sphalerite from different origins. It is indicated that KNN, RF, and NN algorithms are more efficient than other ML methods in the classification tasks. Both PLS-DA and RF methods identify Ge, Mn, In, Sn, Ga and Sb are the most important discriminant elements. The constructed multivariate and ML models are also used to identify sphalerite with uncertain origins. The discrimination results are consistent with the geological setting and results from other geochemical methods. Our study demonstrates that the multivariate and ML models based on sphalerite trace elements can be used as quick screening in mineral exploration of Pb-Zn deposits and related critical metals.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data can be found in the appendix file

## Acknowledgments

This study was funded by the National Key Research and Development Program of China (2021YFC2900300), National Natural Science Foundation of China (42073043, U23A2027, U1812402, 92162323), Provincial Science and Technology Plan Project of Guizhou Province (Qiankehe Foundation-ZK[2023]Key 050), CAS “Light of West China” Program to YMM, CAS Hundred Talents Program to XWH, Special Fund of the State Key Laboratory of Ore Deposit Geochemistry (202101, 202301), and Guizhou Provincial 2020 and 2021 Science and Technology Subsidies (GZ2020SIG, GZ2021SIG). Two anonymous reviewers are thanked for critical reviews.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oregeorev.2024.105887>.

## References

- Abiodun, O.I., Jantan, A., Omolara, A.E., Dada, K.V., Mohamed, N.A., Arshad, H., 2018. State-of-the-art in artificial neural network applications: A survey. *Heliyon* 4, e00938.



- Aitchison, J., 1986. The statistical analysis of compositional data. Chapman and Hall Ltd., London, UK, p. 416.
- Beaudoin, G., 2000. Acicular sphalerite enriched in Ag, Sb, and Cu embedded within color-banded sphalerite from the Kokanee Range, British Columbia, Canada. *Can. Mineral.* 38, 1387–1398.
- Belissont, R., Boiron, M.-C., Luais, B., Cathelineau, M., 2014. LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac-Saint-Salvy deposit (France): Insights into incorporation mechanisms and ore deposition processes. *Geochim. Cosmochim. Acta* 126, 518–540.
- Belissont, R., Munoz, M., Boiron, M.-C., Luais, B., Mathon, O., 2016. Distribution and oxidation state of Ge, Cu and Fe in sphalerite by  $\mu$ -XRF and K-edge  $\mu$ -XANES: insights into Ge incorporation, partitioning and isotopic fractionation. *Geochim. Cosmochim. Acta* 177, 298–314.
- Belyadi, H., Haghighat, A., 2021. Machine Learning Guide for Oil and Gas Using Python: a step-by-step breakdown with data, algorithms, codes, and applications. Gulf Professional Publishing.
- Bouzari, F., Hart, C.J., Bissig, T., Barker, S., 2016. Hydrothermal alteration revealed by apatite luminescence and chemistry: a potential indicator mineral for exploring covered porphyry copper deposits. *Econ. Geol.* 111, 1397–1410.
- Breiman, L., 2001. Random Forests. *Mach. Learn.* 45, 5–32.
- Cook, N.J., Ciobanu, C.L., Pring, A., Skinner, W., Shimizu, M., Danyushevsky, L., Saini-Eidukat, B., Melcher, F., 2009. Trace and minor elements in sphalerite: A LA-ICP-MS study. *Geochim. Cosmochim. Acta* 73, 4761–4791.
- Cooke, D.R., Wilkinson, J.J., Baker, M., Agnew, P., Phillips, J., Chang, Z., Chen, H., Wilkinson, C.C., Inglis, S., Hollings, P.J.E.G., 2020. Using Mineral Chemistry to Aid Exploration: A Case Study from the Resolution Porphyry Cu-Mo Deposit, Arizona. Demšar, J., Curk, T., Erjavec, A., Gorup, Č., Hočevár, T., Milutinović, M., Možina, M., Polajnar, M., Toplak, M., Starč, A., 2013. Orange: data mining toolbox in Python. *J. Mach. Learning Res.* 14, 2349–2353.
- Einaudi, M.T., Hedenquist, J.W., Inan, E.E., 2005. Sulfidation state of fluids in active and extinct hydrothermal systems: Transitions from porphyry to epithermal environments. In: Simmons, S.F., Graham, I. (Eds.), *Volcanic, Geothermal, and Ore-Forming Fluids: Rulers and Witnesses of Processes within the Earth*. Society of Economic Geologists, p. 285–313.
- Eriksson, L., Byrne, T., Johansson, E., Trygg, J., Vikström, C., 2013. Multi- and megavariable data analysis: Basic principles and applications. MKS Umetrics AB, Sweden.
- Frenzel, M., Hirsch, T., Gutzmer, J., 2016. Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type—A meta-analysis. *Ore Geol. Rev.* 76, 52–78.
- Gong, X., Lü, Q., Meng, G., Yan, J., Zhao, J., 2019. Genesis of the Lengshuikeng Pb-Zn Deposit, China: Constraint from In-situ LA-ICP-MS Analyses of Minor and Trace Elements in Sphalerite. ASEG Extended Abstracts 2019, 1–5.
- Guo, F., Wang, Z., Xu, D., Yu, D., Dong, G., Ning, J., Kang, B., Peng, E., 2020. Trace element characteristics of sphalerite in the Lishan Pb-Zn-Cu polymetallic deposit in Hunan Province and the metallogenic implications. *Earth Sci. Front.* 27, 66–81 in Chinese with English abstract.
- Höll, R., Kling, M., Schroll, E., 2007. Metallogenesis of germanium-A review. *Ore Geol. Rev.* 30, 145–180.
- Hron, K., Templ, M., Filzmoser, P., 2010. Imputation of missing values for compositional data using classical and robust methods. *Comput. Stat. Data Anal.* 54, 3095–3107.
- Huang, X.-W., Boutroy, É., Makvandi, S., Beaudoin, G., Corriveau, L., De Toni, A.F., 2019a. Trace element composition of iron oxides from IOCG and IOA deposits: relationship to hydrothermal alteration and deposit subtypes. *Miner. Deposita* 54, 525–552.
- Huang, X.-W., Sappin, A.-A., Boutroy, É., Beaudoin, G., Makvandi, S., 2019b. Trace element composition of igneous and hydrothermal magnetite from porphyry deposits: Relationship to deposit subtypes and magmatic affinity. *Econ. Geol.* 114, 917–952.
- Kramer, O., 2013. K-Nearest Neighbors. In: Kramer, O. (Ed.), *Dimensionality Reduction with Unsupervised Nearest Neighbors*. Springer Berlin Heidelberg, Berlin, Heidelberg, p. 13–23.
- Leach, D.L., Bradley, D., Lewchuk, M.T., Symons, D.T., de Marsily, G., Brannon, J., 2001. Mississippi Valley-type lead-zinc deposits through geological time: implications from recent age-dating research. *Miner. Deposita* 36, 711–740.
- Leng, C., Qi, Y., 2017. Genesis of the Lengshuikeng Ag-Pb-Zn orefield in Jiangxi: Constraint from In-situ LA-ICPMS analyses of minor and trace elements in sphalerite and galena. *Acta Geologica Sinica* 91, 2256–2272 in Chinese with English abstract.
- Li, Q., Jiang, S., 2016. Trace and rare earth element characteristics in Fe-Mn carbonates associated with stratiform Ag-Pb-Zn mineralization from the Lengshuikeng ore district, Jiangxi Province: Implications for their genesis and depositional environment. *J. Earth Sci.* 27, 571–583.
- Li, X.-M., Zhang, Y.-X., Li, Z.-K., Zhao, X.-F., Zuo, R.-G., Xiao, F., Zheng, Y., 2023. Discrimination of Pb-Zn deposit types using sphalerite geochemistry: New insights from machine learning algorithm. *Geosci. Front.* 14, 101580.
- Liu, Y.-F., Qi, H.-W., Bi, X.-W., Hu, R.-Z., Qi, L.-K., Yin, R.-S., Tang, Y.-Y., 2021. Two types of sediment-hosted Pb-Zn deposits in the northern margin of Lanping basin, SW China: Evidence from sphalerite trace elements, carbonate C-O isotopes and molybdenite Re-Os age. *Ore Geol. Rev.* 131, 104016.
- Makvandi, S., Ghasemzadeh-Barvarz, M., Beaudoin, G., Grunsky, E.C., McClenaghan, M. B., Duchesne, C., 2016a. Principal component analysis of magnetite composition from volcanogenic massive sulfide deposits: Case studies from the Izok Lake (Nunavut, Canada) and Halfmile Lake (New Brunswick, Canada) deposits. *Ore Geol. Rev.* 72, 60–85.
- Makvandi, S., Ghasemzadeh-Barvarz, M., Beaudoin, G., Grunsky, E.C., McClenaghan, M. B., Duchesne, C., Boutroy, E., 2016b. Partial least squares-discriminant analysis of trace element compositions of magnetite from various VMS deposit subtypes: Application to mineral exploration. *Ore Geol. Rev.* 78, 388–408.
- Metz, C.E., 1978. Basic principles of ROC analysis. *Seminars in Nuclear Medicine* 8, 283–298.
- Miranda, A.C.R., Beaudoin, G., Rottier, B., 2022. Scheelite chemistry from skarn systems: implications for ore-forming processes and mineral exploration. *Miner. Deposita* 1–29.
- Mukherjee, I., Large, R., 2017. Application of pyrite trace element chemistry to exploration for SEDEX style Zn-Pb deposits: McArthur Basin, Northern Territory, Australia. *Ore Geol. Rev.* 81, 1249–1270.
- Pisiak, L.K., Canil, D., Lacourse, T., Plouffe, A., Ferbey, T., 2017. Magnetite as an indicator mineral in the exploration of porphyry deposits: A case study in till near the Mount Polley Cu-Au deposit, British Columbia, Canada. *Econ. Geol.* 112, 919–940.
- Pisner, D.A., Schnyer, D.M., 2020. Chapter 6 - Support vector machine. In: Mechelli, A., Vieira, S. (Eds.), *Machine Learning*. Academic Press, pp. 101–121.
- Rish, I., 2001. An empirical study of the naive Bayes classifier. *IJCAI 2001 workshop on empirical methods in artificial intelligence*, p. 41–46.
- Sciuba, M., Beaudoin, G., Makvandi, S., 2021. Chemical composition of tourmaline in orogenic gold deposits. *Miner. Deposita* 56, 537–560.
- Stoiber, R.E., 1940. Minor elements in sphalerite. *Econ. Geol.* 35, 501–519.
- Sun, G.-T., Zhou, J.-X., 2022. Application of Machine Learning Algorithms to Classification of Pb-Zn Deposit Types Using LA-ICP-MS Data of Sphalerite. *Minerals* 12, 1–16.
- Van der Maaten, L., Hinton, G., 2008. Visualizing data using t-SNE. *J. Machine Learning Res.* 9, 2579–2605.
- Wang, C., Zhang, D., Wu, G., Santosh, M., Zhang, J., Xu, Y., Zhang, Y., 2014. Geological and isotopic evidence for magmatic-hydrothermal origin of the Ag-Pb-Zn deposits in the Lengshuikeng District, east-central China. *Miner. Deposita* 49, 733–749.
- Whitten, E.H.T., 1995. Open and closed compositional data in petrology. *Math. Geol.* 27, 789–806.
- Wold, S., Esbensen, K., Geladi, P., 1987. Principal component analysis. *Chemometrics and Intelligent Laboratory Systems* 2, 37–52.
- Ye, L., Cook, N.J., Ciobanu, C.L., Yiping, L., Qian, Z., Tiegeng, L., Wei, G., Yulong, Y., Danyushevskiy, L., 2011. Trace and minor elements in sphalerite from base metal deposits in South China: a LA-ICPMS study. *Ore Geol. Rev.* 39, 188–217.
- Yu, D., Xu, D., Wang, Z., Xu, K., Huang, Q., Zou, S., Zhao, Z., Deng, T., 2021. Trace element geochemistry and O-S-Pb-He-Ar isotopic systematics of the Lishan Pb-Zn-Cu hydrothermal deposit, NE Hunan, South China. *Ore Geol. Rev.* 133, 104091.
- Zhang, Q., 1987. Trace elements in galena and sphalerite and their geochemical significance in distinguishing the genetic types of Pb-Zn ore deposits. *Chin. J. Geochem.* 6, 177–190.
- Zhang, J., Shao, Y., Liu, Z., Chen, K., 2022. Sphalerite as a record of metallogenic information using multivariate statistical analysis: Constraints from trace element geochemistry. *J. Geochem. Explor.* 232, 106883.